Universal Internet Access at What Cost? Predicting Drivers of Funding Allocation under the Broadband Equity, Access, and Deployment (BEAD) Program

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Abstract

The Broadband Equity, Access, and Deployment (BEAD) program is the largest single federal investment in broadband infrastructure in United States history. The program, administered by the National Telecommunications and Information Administration (NTIA), allocates $42.5 billion among states and territories to expand high-speed internet access to unserved and underserved locations. Each state, in turn, was free to use its own methodology in choosing winning bids but was constrained by local factors such as competitive intensity and terrain. The per-location BEAD funding varies widely, but little is known about what drives this variation.

In this paper, we analyzed BEAD final proposal data, selected a set of cost features, and evaluated their predictive power using a series of machine learning models. Our tuned Random Forest explained approximately 71.9% of the variance in per-location funding, compared to 54% for linear approaches, demonstrating that deployment specifications and funding have a nonlinear relationship. Fiber miles per location and dominant technology type are the strongest predictors.

CCS Concepts

• **Social and professional topics** → **Governmental regulations**; • **Networks** → **Network economics**; *Public Internet*; • **Applied computing** → *Decision analysis.*

Keywords

Broadband Access, Telecommunications Policy, Predictive Modeling, Infrastructure Funding Allocation, Latent Infrastructure

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1 Introduction

The Broadband Equity, Access, and Deployment (BEAD) program, passed in 2021, has a budget of $42.45 billion, making it the largest single federal investment in broadband [1, 19]. As broadband infrastructure underpins digital transformation across sectors, understanding funding decisions has implications beyond telecommunications policy [21]. Ensuring fair and transparent BEAD funding decisions should not only be an economic and policy question but also directly relate to universal and reliable access to a critical service.

BEAD funding distribution can be divided into two main stages: NTIA allocated funding to Eligible Entities[1], and each Eligible Entity conducted subgrantee selection through their final proposal. §2.1 illustrates how NTIA determines funding allocation for high-cost areas—10% of BEAD funding—using its cost model with remoteness, population density, topography, and poverty levels. States then distribute these funds as subgrants to internet service providers (ISPs) willing to build broadband to a group of locations. BEAD makes funding available to build broadband that meets the following criteria: at least 100 megabits per second (Mbps) download and 20 Mbps upload speed to unserved and underserved locations ("100/20"). Unserved locations lack 25/3 Mbps broadband; underserved locations lack 100/20 Mbps access. Only terrestrial broadband technologies such as fiber, hybrid fiber–coaxial (HFC) cable, and fixed wireless are considered to determine whether a location is unserved or underserved. Despite BEAD's large scale, little empirical work has examined what drives variation in average per-location funding (APLF) during the subgrantee selection stage. Drawing inspiration from NTIA's cost model for funding allocation, past efforts such as Bauer et al. [4] used multivariate linear regression modeling to

[1]Eligible Entities include all 50 U.S. states, the District of Columbia, and five U.S. territories. Due to their unique nature, we excluded territories from our analysis.

predict APLF using factors such as terrain ruggedness and rural area percent and found a small R^2 of 0.189.

Unlike how NTIA's cost model informs its funding allocation, variation in state-level subgrantee selection can be plausibly attributed to technology type, geography, and market structure but remains largely empirically untested [15, 16].

Using project- and location-level data from the Final Proposal (FP) from 49 states and the District of Columbia and the FCC's Broadband Data Collection (BDC), we fill this gap by testing a series of predictive models to identify factors that best explain variation in APLF among BEAD projects. Our contributions are as follows:

We present **one of the first predictive models on average per-location funding in BEAD FP**. Our model handles nonlinear relationships and explains approximately 71% of the variation in average per-location funding among BEAD projects. We further characterize unexplained variance and attribute it to factors such as between-state variance.[2]

We quantify **the effect of market competition, awardee incumbent footprint, and geographic factors** on funding variation observed in subgrantee selection, which has been theorized in works such as Lehr [16] and Jamison [15] but has yet to be empirically modeled.

2 Background and Motivation

2.1 Overview of the BEAD Program

BEAD funding distribution occurs on two levels: NTIA to Eligible Entities, and Eligible Entities to subgrantees[3]. In the first stage, the National Telecommunications and Information Administration (NTIA) distributed funding among 56 eligible states, territories, and the District of Columbia ("Eligible Entities") according to the allocation priorities listed in the Infrastructure Investment and Jobs Act (IIJA) [1, 19]. The three components include

(1) a baseline cost of $100 million for each state and $25 million for each territory,
(2) a weighted distribution based on each state's number of unserved locations[4] divided by the nationwide total, and
(3) 10 percent distributed based on each state's "high-cost" areas divided by the nationwide total [19].

Of the three components, the second depends on the number of unserved locations from the Broadband Data Collection (BDC) administered by the Federal Communications Commission (FCC) [10, 15]. First released in 2022 on a biannual basis, the BDC is the first nationwide location-level census of broadband availability, despite challenges such as self-reported coverage by internet service providers (ISPs) [11]. High-cost funding was therefore allocated using a combination of factors.

The third component depends on "high-cost" areas, which are census block groups with at least 80% unserved locations. These are determined using a cost model incorporating remoteness, population density, topography, and poverty levels [1, 12, 19].

In the second stage, Eligible Entities distributed their allotted NTIA funding to subgrantees, following BEAD's Notice of Funding Opportunity (NOFO) [27] and the June 2025 "Benefit of the Bargain" Restructuring Policy Notice (RPN) [28]. Subject to NTIA oversight, states could decide how to divide their list of locations into project areas and which technologies to deploy in each location [19].

Some state broadband offices evaluate subgrantees' bids using CostQuest's biannual cost model. Specifically, BEAD's NOFO formerly contained an "Extremely High Cost per Location Threshold" (EHCPLT): if a Fiber-to-the-Premises (FTTP) solution exceeds this threshold, then states can accept non-FTTP bids that meet BEAD speed requirements [27]. One CostQuest analysis of an unnamed state illustrated the EHCPLT's intent to prevent a disproportionate share of total investment in serving a small fraction of locations: 2.5% of locations required 28.1% of the total fiber investment [2]. As of June 2025, the RPN no longer required states to establish an EHCPLT. Instead, NTIA reserved the right to reject excessively costly projects on a case-by-case basis and restructured subgrantee selection to prioritize lowest cost per location, opening competition to all qualifying technologies including FW and LEO [28].

This paper focuses on the subgrantee selection stage, investigating predictors that may explain the differences in Eligible Entities' selection of awarded ISPs.

2.2 Current BEAD Funding Debates

The existing literature on BEAD funding allocation centers on several interconnected debates: technology-specific deployment costs, the ambiguous relationship between market competition and subsidy efficiency, and the influence of auction design and state-level process variation on observed outcomes.

On technology-specific deployment costs, Jamison [15] critiqued NTIA's previous prioritization of fiber-to-the-premise (FTTP) solutions even where cheaper technologies may suffice, framing this as a policy-level cost driver rather than a market outcome [27]. Lehr [16] extended the argument that fiber mandates raise costs and delay broadband benefits, especially in areas where fixed wireless would be cheaper, and that costs rise steeply as population density falls. Early industry cost modeling also reflected similar tensions. Cartesian and ACA Connects [9] projected that deploying fiber to all eligible locations would cost between $45-$56 billion, with per-location costs rising steeply in the highest-cost deciles, highlighting technology and geographical constraints of cost drivers. Since the RPN in June 2025 removed the fiber preference, subgrantee selection processes adapted to balance cost between traditional fiber builds and alternatives like fixed wireless and satellite [28].

On market concentration, little is known about the relationship between provider competition and subsidy efficiency [15]. A concentrated market with fewer larger providers may benefit from economies of scale, yet may enable them to capture excess subsidy. Since market concentration is not readily observable in available data, we derived binary labels for each BEAD-funded location by checking if there exists any qualifying competitor, or any ISP other than the awardee that meets the 100/20 threshold and is a terrestrial provider, as a way to approximate competition. Other features

[2] We made our repository public at https://github.com/yusunnyf/bead_funding_predictor
[3] See Appendix A for glossary
[4] **Served**: meets 100 Mbps download/20 Mbps upload and is a reliable broadband service
Unserved: under 25 Mbps download/3 Mbps upload
Underserved: "In-between;" i.e., exceeds unserved threshold but doesn't meet served threshold

in §4.2, such as nearest served location, explain the snapshot of provider availability in funded locations.

On state-to-state variations in subgrantee selection, Rosston and Wallsten [23, 24] highlighted the tension between each Eligible Entity's spending efficiency and financial sustainability. According to the June 2025 RPN, NTIA determines the conditions under which states can retain unspent funds [28]. Furthermore, within each Eligible Entity, auction-based bidding, which sometimes requires multiple rounds, may push ISPs to reveal their minimum subsidy needs [15]. Since ISP footprint, terrain distribution, and governance differ by state, a core challenge may be that a predictive model built on all available projects may not generalize equally well to states represented in the training data versus states that were not, an adjacent topic we explored briefly in §5.3.

3 Data and Exploration

3.1 Overview of Datasets

Output variable: average per-location funding (APLF). For subgrantee selection, each Eligible Entity submitted a Final Proposal and attached dataset for NTIA approval [20]. As of August 26th, 2026, the NTIA approved final proposals from all 56 Eligible Entities. Based on data availability, our dataset contains 49 states. We corroborated each state's final proposal with NTIA-published figures. Since some state broadband offices have yet to publish their approved final proposal, we summarized the differences in total BEAD support, total subgrantee match, and total number of locations by technology in Appendix B.2.[5]

Our output variable, *average per location funding* (APLF), is calculated by dividing project-level `bead_support` by the number of locations within a project. Since states are not required to disclose funding distributed across locations in a given project, our target variable is the next-best estimate for project-level average funding. Some states bundled project areas used for bidding into a smaller number of FP project areas, thus obscuring regional differences in competition or cost.

Input variables. Our candidate input variables belong to four categories. **Project-level** data come directly from each state's FP, including a deployment project, a location, a community anchor institutions (CAI) table, and a subgrantee table. The subgrantee table helped filter existing providers via their FCC Registration Number (FRN), whereas the project, location, and CAI tables were used directly in deriving features.

We turned to the National Broadband Map (NBM), part of the FCC's BDC [10], to derive **competition proxies**, such as the presence of qualifying competitors at a given location, and **awardee ISP footprint**, such as the average percentage of census block it served pre-BEAD. During the "Benefit of the Bargain" round, Eligible Entities were asked to submit an updated FP to the NTIA by September 2025; thus, we selected the December 2024 NBM as the snapshot Eligible Entities likely referenced in drafting their FPs [26]. Some features required longitude and latitude for calculation, which the Broadband Serviceable Location (BSL) Fabric from CostQuest made possible [3].[6]

[5]NTIA-published figures can be found at its website
[6]Columbia University has a Tier 4 Research License with CostQuest

To understand how NTIA cost factors may affect more granular funding decisions, we utilized datasets published by the U.S. Census Bureau and the Economic Research Service of the U.S. Department of Agriculture (USDA) to aggregate **county-level** demographic and geographic features. Table 1 summarizes data sources and acronyms.

Table 1: Datasets, acronyms, and corresponding publication dates for input variables

Dataset	Acronym	Year
FP: BEAD-funded locations	L	Aug 2026
FP: Cojoined Location-CAI	LC	Aug 2026
FP: Deployment Projects	P	Aug 2026
BDC: National Broadband Map	NBM	Dec 2024
CostQuest: BSL Fabric, release 6 [3]	BSL	Dec 2024
USDA: Rural-Urban Continuum Codes [31]	RUCC	2023
ACS: Pop. with poverty status [30]	ACS	2020-24

3.2 Exploratory Data Analysis

Before modeling, exploratory data analysis (EDA) characterizes the distribution of BEAD funding at a project- and location-level as well as state-level variations in BEAD funding and project counts. We initially assessed the data for normality, an important consideration for parametric hypothesis testing such as t-tests. Because the raw monetary variables were strongly right-skewed, we applied a log1p transformation to BEAD support and funding per location. After the log transformation, as shown in Figure 1, the transformed distributions are closer to a normal distribution.

The QQ plots in Figure 2 reveal a zero-inflation floor and multimodal structure. These deviations suggest caution when applying normality-based hypothesis tests and indicate that standard Ordinary Least Squares (OLS) regression may not fully capture the structure of the data.

At the state level, Figure 3 highlights geographic variation in average funding per location, with gray states indicating unavailable BEAD data. North Dakota exhibits the highest average funding per location, exceeding $20,000, while Alaska also shows comparatively high funding levels. These differences suggest that deployment costs may vary considerably across states due to geographic and infrastructural factors such as terrain, population density, remoteness, and existing broadband infrastructure. We examine the relationship between these factors and funding allocation more systematically in our subsequent predictive modeling.

The ranked view in Figure 4 (see Appendix C for the full ranking) is shown for selected states, which reflect the unequal sample sizes present in the data rather than an idealized distribution. The intervals widen alongside the statewide mean: high-cost states, such as North Dakota, exhibit greater within-state dispersion, and their high means rest on fewer, more variable projects.

Average project size in Figure 5 reveals substantial variation in how BEAD projects are structured across states (see Appendix C for the complete state-level breakdown). Ohio has the largest average project size at roughly 3,600 locations per project, followed by California at about 3,300, while Arizona, Montana, and New York also exceed 2,000 locations per project. At the opposite end, Eligible

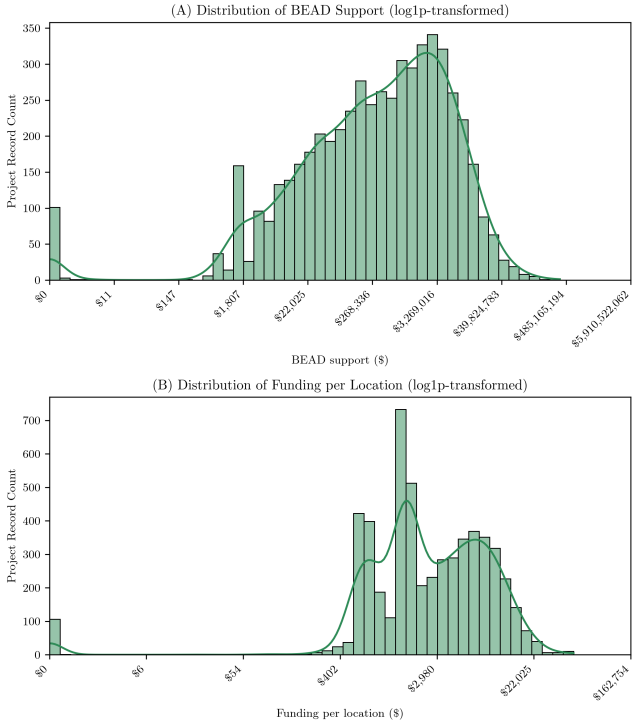


Figure 1: Log-transformed distribution of (A) total BEAD support and (B) average per-location funding for all projects.

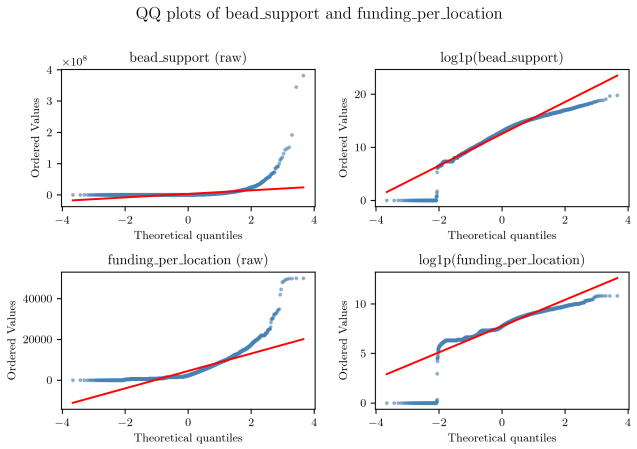


Figure 2: QQ plots for BEAD support and funding per location.

Entities such as the District of Columbia, Nebraska, Connecticut, and Indiana have projects covering only a small number of locations on average. This pattern suggests that project counts alone can be misleading: a state with many projects may simply divide its eligible locations into smaller project areas, whereas another may aggregate thousands of locations into relatively few awards. As Eligible Entities have discretion in defining project areas, cross-state

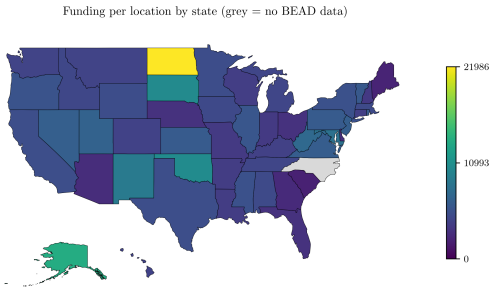


Figure 3: Choropleth of funding by state with North Dakota as an extreme outlier

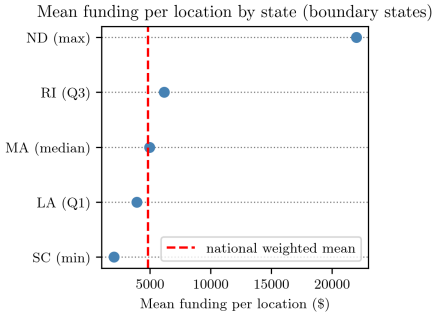


Figure 4: Mean BEAD funding per location by state. Full ranking in Figure 8, Appendix C.

differences in project volume may therefore reflect administrative granularity as much as underlying deployment scale or cost.

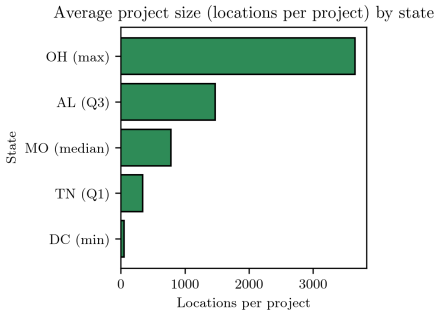


Figure 5: Average project size (locations per project) per state. Full breakdown in Figure 9, Appendix C.

Taken together, the exploratory analysis repeatedly points towards the same structural divide. The bimodality in the log-transformed target, the geographic dispersion in per-location funding, and the different project sizes suggest underlying differences in the deployment of subsidized broadband that warrant statistical scrutiny. In the next section, we conduct nonparametric hypothesis testing to investigate whether factors outside of deployment cost drive funding differences.

3.3 Hypothesis Testing

We formalized a set of hypotheses to test drivers of funding-per-location. Due to the right-skewed nature of the variable, we rely on non-parametric rank-based tests that do not assume normality and are robust against heavy tails. We report effect sizes alongside p-values, since the large sample size ($n \approx 5{,}400$ projects) renders statistical significance easy to attain and effect magnitude the more informative quantity. Table 2 summarizes the tests conducted.

Table 2: Summary of hypothesis tests. All tests are rank-based and make no normality assumption. Effect sizes: r for Mann-Whitney, ε^2 for Kruskal-Wallis, ρ for Spearman.

#	Hypothesis	Effect	Supported
H1	APLF differs by technology types	$\varepsilon^2 = 0.49$	Y
H2	Fiber APLF costs more than FW	$r = +0.51$	Y
H3	Buried fiber costs more than aerial	$r = -0.01$	N
H4	APLF rises with miles per location	$\rho = +0.72$	Y
H5	Wired costs more than non-wired	$r = +0.77$	Y
H6	Governor incumbency affects APLF	$r = +0.38$	N

H1: Technology is the dominant categorical driver. A Kruskal-Wallis test comparing average per-location funding based on each project's most prevalent technology yields significant results and a large effect size ($H = 2644.4$, $p \ll 0.001$, $\varepsilon^2 = 0.49$). Roughly a third of the rank variation is attributable to technology type alone, with median funding per location ranging from \$1,500 for LEO satellite to \$1,709–\$2,544 across fixed-wireless types, compared with \$5,531 for HFC and \$5,812 for fiber.

H2: Fiber costs more than fixed wireless (FW). A one-sided Mann-Whitney test supports that fiber projects[7] get higher funding than fixed wireless ones ($p \ll 0.001$, $r = +0.51$), with median fiber funding roughly 2.75× that of fixed wireless (\$5,812 vs. \$2,116). This supports fiber's known higher construction capital requirement.

H3: Buried and aerial fiber do not differ per location. We find no significant difference in funding per location between buried-dominant and aerial-dominant fiber projects (two-sided Mann–Whitney $p = 0.67$, $r = -0.01$; median buried \$5,672 versus aerial \$5,937). Of the 2,712 fiber projects, 1,566 (57.7%) are buried-dominant, 951 (35.1%) are aerial-dominant, and 195 (7.2%) have equal shares. Half of these projects (1,359; 50.1%) combine both placement methods, with buried fiber accounting for an average of 42.6% of miles among mixed projects. These results do not imply equal construction costs per mile. They only indicate that project-level funding per location does not differ between the two dominant-placement groups.

H4: Cost rises sharply with build requirements. Spearman's ρ between APLF and fiber miles per location is strongly positive ($\rho = +0.72$, $p \ll 0.001$), and Pearson correlation on the same variables produced a coefficient close to zero ($r = 0.05$), confirming that the miles-to-funding relationship is monotonic but nonlinear, which motivates our usage of Random Forest (See Appendix C for scatterplot).

H5: The cost signal is fundamentally wired-versus-wireless. To test whether fiber's high costs reflect something broader, we merged HFC cable and fiber into a single wired category and re-tested. Cable, though a tiny share of projects (~72 projects, 1.3%), is as expensive as fiber per location (Median \$5,805 vs. \$5,531, $p = 0.16$). The median funding per location merged wired group significantly exceeds fixed wireless (\$5,805 vs. \$2,116, $p \ll 0.001$, $r = +0.51$), LEO (\$5,805 vs. \$1,500, $p \ll 0.001$, $r = +0.87$) and all non-wired technologies combined (\$5,805 vs. \$1,500, $p \ll 0.001$, $r = +0.77$). Among wireless technologies, fixed wireless is also significantly more expensive than LEO, with median costs of \$2,116 versus \$1,500 ($p \ll 0.001$, $r = +0.47$). H2's fiber-versus-wireless result is thus best understood as one symptom of a broader wired-versus-wireless divide, where the dominant cost boundary in BEAD deployment separates physical wire infrastructure from wireless and satellite delivery.

H6: Governor incumbency has no detectable effect on funding. To understand administrative capacity and political influence, we tested whether states with a longer-tenured (incumbent) governor received systematically different funding than first-term-governor states. Analysis at the state level (38 incumbents, 12 first-term) revealed that the difference in median funding per location is not significant ($p = 0.13$, $r = +0.29$). Incumbent-governor states do skew higher-funding (\$4,191 vs. \$2,166 median-of-medians), but with a small sample, this gap falls within random variation and could be confounded by geography. We report this null to illustrate the importance of matching the test's unit of analysis to the level at which the treatment varies.

The hypothesis tests converge on a coherent account of what drives per-location cost in BEAD deployment. The strongest signal is build intensity, measured as fiber miles per location (H4). Closely related and nearly as dominant is a divide in infrastructure requirement, or whether a project delivers service over wired plant or over wireless technologies (H1, H2, H5), accounting for roughly half the rank variation in cost. Fiber-heavy projects require extensive mileage per location, so build intensity and technology type jointly capture the demands of reaching dispersed, hard-to-serve areas.

Two negative results are equally insightful. The lack of difference between buried and aerial miles (H3) indicates that per-location economics are governed by distance regardless of deployment method. Absence of a governor-incumbency effect (H6) cautions against reading political explanations into variations that geography may already account for. Together, these findings establish the empirical priors that motivate feature choices in §4.2: fiber miles per location and technology type are likely to carry the most predictive weights.

4 Predictive Modeling

Random Forest, our choice of model, is an ensemble machine learning algorithm that trains a collection of independent decision trees and averages their predictions for a more robust estimate. It assumes no linear relationships and can thus model nonlinear relationships with both continuous and categorical features. §4.2 details our feature selection, and §4.3 discusses hyperparameter tuning.

[7] Throughout this section, "x projects" refers to projects with the majority ($\geq 50\%$) of their locations assigned to tech x.

4.1 Mathematical Formulation of Model

Consider a Random Forest (RF) model. Let the training dataset be

$$D = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n},$$

where $\mathbf{x}_i \in \mathbb{R}^p$ is a p-length vector of predictor variables for project i, and y_i is the log-transformed funding per location. The model consists of T regression trees, each trained on a bootstrap sample D_t by sampling n observations from D with replacement [6].

At each node, a random subset of m features ($m < p$) is selected. Among these features, the split is chosen to minimize the mean squared error (MSE) of the resulting child nodes,

$$(j^*, s^*) = \arg\min_{j \in \mathcal{M}, s} \left(\frac{|S_L|}{|S|} \mathrm{MSE}(S_L) + \frac{|S_R|}{|S|} \mathrm{MSE}(S_R) \right),$$

where $\mathcal{M}$ is the selected subset, S is the set of observations reaching the current node, and S_L and S_R denote the left and right child nodes split by threshold s. The mean squared error for S is

$$\mathrm{MSE}(S) = \frac{1}{|S|} \sum_{i \in S} (y_i - \bar{y}_S)^2,$$

where $\bar{y}_S$ is the mean response of observations within that node.

For a new observation $\mathbf{x}$, the Random Forest prediction is obtained by averaging the predictions from all T regression trees,

$$\hat{f}(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^{T} f_t(\mathbf{x}),$$

where $f_t(\mathbf{x})$ denotes the prediction produced by the tth tree. Averaging across independently bootstrapped trees reduces the estimator's variance while preserving its ability to model nonlinear relationships, improving predictive performance and reducing overfitting compared to an individual decision tree [6].

4.2 Feature Engineering

Drawing on datasets in §3.1, we selected several features to model Eligible Entities' funding decisions. Table 3 details the input variables, respective sources, and categories. Refer to Table 13 in the Appendix for more information. Feature ablations are shown in Table 12 in Appendix D.3.

Our input variables span four main categories: project-level features, competition proxies, awardee ISP footprint, and county-level features. **Project-level features** are direct engineering metrics from the FP. To differentiate broadband technology, we created a coarser categorization based on physical deployment requirements, although the NBM has more fine-grained technology codes.[8] Technology codes don't encode any ordinality; thus, we first one-hot encoded technology assignment at the location level before aggregating it by project to get the percentage of locations assigned to each technology per project. We also computed average fiber miles per location, ratio of aerial fiber miles, and ratio of CAI assignments.

For **competition proxies**, we computed the fraction of locations with qualifying competitors[9] as of December 2024, aggregating location-level binary flags to the project level. As noted in Table 3, we only consider competitors with terrestrial infrastructure since essentially all locations have satellite coverage.

We also investigated the **awardee ISP's footprint** as of December 2024, including its average census block-level coverage across funded locations and the percentage of funded locations with its presence. For each funded location, we computed the distance to the nearest served neighbor using the Haversine Ball Tree [8, 25]. Distance to the nearest own-served location approximates construction and extension costs, while distance to the nearest served location of any ISP captures terrain remoteness more generally.

County-level demographic and geographic variables help test how features NTIA's cost model alluded to may also have an impact on subgrantee selection. Using Analysis of Variance (ANOVA), we filtered out features whose variance is attributable to state membership to prevent data leakage. More details on motivation for each feature and relevant literature can be found in Appendix D.4.

Our output variable is the log-transformed *average per-location funding* (log-APLF), denoted by $y^* = \log(1 + y)$, where y is the raw APLF. Normalizing by the number of funded locations removes the effect of project size, allowing the analysis to focus on the underlying cost drivers of broadband deployment. We trimmed the 2.5th and 97.5th percentiles of APLF to reduce outlier influence (results on the untrimmed data are in §5.2).

4.3 Model Configuration

To predict log-transformed BEAD funding per location, we trained a single Random Forest regressor on $n = 5,188$ projects, after trimming the outliers. We used a combination of RandomizedSearchCV and manual tuning to find optimal model parameters [5]. Our tuned model consists of 1,000 independent estimators, each with a max depth of 20, a subset of at most 30% of features, a minimum sample split of 2, and at least 1 sample per leaf.

With a fixed random seed for reproducibility, we used a 70/30 train-test split and a 5-fold cross-validation with shuffled folds to ensure consistency and stability across all runs, including ablation studies in §5.

4.4 Performance

The tuned model achieves a 5-fold cross-validated R^2 of 0.6920 ± 0.0145, showing consistent performance across folds. The model's test $R^2 = 0.7192$, explaining 71.9% of the variance in APLF across unseen projects. The low standard deviation across folds (± 0.0145) indicates consistent predictive performance not driven by a particular train-test split. In dollar terms, the model predicts BEAD funding per location with a root mean squared error (RMSE) of $\$2,645$ and a mean absolute error (MAE) of $\$1,546$. The MAE tells us that predictions deviate from observed funding amounts by roughly $\$1,546$ on average. Since RMSE penalizes larger residuals more heavily, the higher RMSE suggests that there is a relatively small number of larger prediction errors.

Figures 6A and B show a trend: as APLF increases, predictions become more error-prone. One likely explanation is that projects with a higher average funding are inherently more heterogeneous and difficult to model.

[8] As mentioned in Table 3, we grouped {70, 71, 72} as FW more generally.
[9] As mentioned in Table 3, a qualifying competitor is any ISP other than the awardee that meets the 100/20 threshold and is a terrestrial provider

Table 3: Input variables, data sources, and description

Category	Variable	Description	Data Source
Project-level	`n_total`	row (location and CAI) count in project	LC
	`frac_tech`	rows assigned to $x^{\dagger}$ divided by `n_total`	LC
	`miles_per_loc`	total fiber miles divided by `n_locations`	L, P
	`pct_aerial`	total aerial fiber miles divided by total fiber miles	P
	`pct_cai`	count of CAI divided by `n_total`	LC
Competition	`frac_q_comp`	row count with qualifying competitor(s)‡ divided by `n_total`	LC, NBM
	`frac_q_wired`	row count with qualifying wired (40, 50) competitor(s) divided by `n_total`	LC, NBM
Awardee ISP	`coverage`	percentage of census block covered by ISP for each loc, averaged	LC, NBM
	`frac_presence`	row count with awardee ISP divided by `n_total`	LC, NBM
	`nearest_own_mi`	distance (in miles) to the nearest served location with same ISP, averaged	BSL, LC, NBM
	`nearest_any_mi`	distance (in miles) to the nearest served location with any ISP, averaged	BSL, LC, NBM
	`no_prior_presence`	row count with no nearest served location with same ISP divided by `n_total`	LC, NBM
County-level	`rur_popden`	loc-weighted mean of persons per miles^2 of rural land area across counties	LC, C
	`cou_popden`	loc-weighted mean of persons per miles^2 of rural land across counties	LC, C
	`rucc_code`	loc-weighted mean of Rural-Urban Continuum Code (RUCC) across counties	LC, RUCC
	`poverty_rate`	loc-weighted mean of poverty rate across counties	LC, ACS

† Tech type $x \in \{40, 50, 61, 71\}$, 40 = Cable, 50 = Fiber, 61 = LEO Satellite; 71 = Fixed Wireless
‡ Competitor: any ISP that is not the awardee
Qualifying competitor: any ISP other than the awardee that meets the 100/20 threshold and is a terrestrial provider (i.e., not 61)

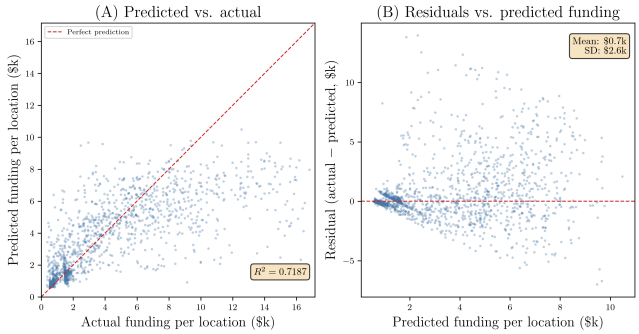


Figure 6: (A) Predicted vs. actual values of the tuned RF model. (B) Residual plot for the predicted dollars of funding per location. Dots further away from the horizontal line at 0 indicate more error.

4.5 Interpreting Feature Importance and Effect

Table 4 summarizes the feature importance of our model using two methods: Random Forest Gini importance measures the overall contribution of each feature to reducing prediction error across all trees in the forest, and SHapley Additive exPlanations (SHAP) quantifies how each feature increases or decreases the predicted funding for every individual observation while providing a consistent global ranking of feature importance. Figure 7 shows the SHAP feature effects. Each dot represents one project. The horizontal position indicates the feature's direction and extent of impact on that prediction, whereas the color encodes the feature's raw value.

Table 4: Normalized feature importance from RF Gini and SHAP; full table in Appendix D.2

Feature	RF Gini	SHAP
`miles_per_loc`	0.241	0.275
`frac_tech_61 (LEO)`	0.099	0.174
`frac_tech_50 (Fiber)`	0.126	0.142
`nearest_own_mi`	0.063	0.061
`rur_popden`	0.049	0.045
`frac_presence`	0.039	0.044
`nearest_any_mi`	0.045	0.043
`isp_coverage`	0.053	0.034
`poverty_rate`	0.037	0.028
`n_total`	0.048	0.023

In Table 4, miles per location is shown to be the most influential feature in the model, accounting for around 20% of the model's total importance and suggesting physical distance required to reach each fiber location is the primary predictor of BEAD funding. Consistent with the Gini importance results, "miles per location" has the largest impact on funding predictions in terms of SHAP values. With high values (red) clustered on the right and low values (blue) clustered on the left in Figure 7, miles per location shows a positive monotonic effect on APLF, indicating that longer miles per location substantially increase predicted APLF.

Technology composition also exhibits a noticeable effect, with the fraction of LEO and fiber locations per project being runner-ups in importance. Interestingly, the two features exhibit near-opposite effects in Figure 7: LEO-designated projects—those where virtually all awarded locations are assigned LEO service—receive substantially lower APLF, while projects composed predominantly of fiber

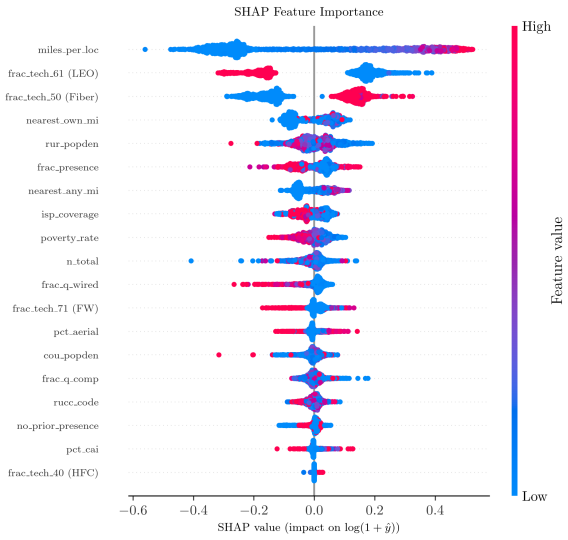


Figure 7: SHAP values for Baseline Model

locations receive higher APLF. Together, this echoes the intent of the June 2025 RPN: with the EHCPLT replaced by a cost threshold above which bids could not be selected, lower-cost technologies such as FW and LEO were more likely to win, driving down APLF [28].

5 Ablation Studies

5.1 Model Comparison

To determine which best suits predicting APLF, we trained both linear and nonlinear models. Table 5 summarizes ablation results.

Pursuant to Bauer et al. [4], we first fitted linear models to predict APLF. The authors used the same target variable with synonymous features, such as rural area characterization and poverty rate, making the R^2 comparison meaningful despite differences in model specification and sample scope. Thus, we first tested an Ordinary Least Squares (OLS) model and found multicollinearity among the feature set. To address this, we tried Ridge Regression with L2 regularization, which adds a penalty term proportional to the sum of squared coefficients and shrinks the regression coefficients. Similarly, we used Lasso Regression with L1 regularization, which adds a penalty term proportional to the sum of absolute coefficients and effectively shrinks less important coefficients to zero, affirming miles per location as a key predictor.

After observing that linear models, even with measures to handle multicollinearity, had relatively high RMSE and MAE, we tested three tree-based methods — Random Forest (RF), Gradient Boosted Regression Trees (GBR), and XGBoost — on the same feature set. Each builds separate decision trees, but combines trees differently. RF trains trees independently on bootstrap samples and averages their predictions, whereas GBR and XGBoost build sequential trees that each correct the previous one's residuals to reduce bias.

As shown in Table 5, tuned RF achieved the highest CV R^2 of 0.7116, with XGBoost as a close second. Because of its robustness to overfitting, low sensitivity to multicollinearity, and ability to capture non-linear relationships, we chose RF as our final model. While the negative R^2 values for all linear models all point towards severe multicollinearity, Random Forest models increased that number to around 71.9%. This is due to the following two structural reasons. One, miles per location has a highly nonlinear relationship with APLF, as shown in H2 of §3.3. Second, deployment cost scales with miles for fiber differently than for fixed wireless or LEO satellite due to different infrastructure requirements.

5.2 Outlier Trimming

Our main model is trained on n = 5,188 projects after trimming the 2.5th and 97.5th percentiles of APLF. To check how much this choice affects our results, we refit the same tuned Random Forest on the untrimmed data (n = 5,462) using the same 19 features. Table 6 compares the two settings.

Table 6: Effect of outlier trimming on the tuned Random Forest. R^2 is on the log-transformed target and CV R^2 is mean ± SD across 5 folds. RMSE and MAE are in dollars on each setting's own test set and are not directly comparable.

	Trimmed (main)	Untrimmed
n (projects)	5,188	5,462
Test R^2	0.7187	0.4494
CV R^2	0.6920 ± 0.0145	0.4987 ± 0.0648
RMSE ($)	2,645	3,831
MAE ($)	1,546	1,946

Without trimming, the same feature set explains substantially less variance (test R^2 = 0.449, CV R^2 = 0.499, versus 0.719 and 0.692 in the trimmed model), and dollar errors rise (RMSE $3,831 vs. $2,645; MAE $1,946 vs. $1,546). This suggests that projects in the extreme tails of APLF are poorly explained by our features, so our headline R^2 should be read as applying to the central 95% of projects rather than all projects. The wider gap between test and CV R^2 in the untrimmed setting also indicates that a single split is sensitive to where the extreme projects land. Because dollar-scale errors are computed on different test sets, we treat log-scale R^2 as the primary basis for comparison.

5.3 State-Stratified Modeling

To understand how well the model's predictive power transfers across state boundaries, we evaluated two ablations: (1) cross-state, where the model was trained on a subset of states and tested on a held-out set, and (2) within-state, where the model was trained and tested within a single state.

(1) Cross-state modeling. Projects within the same state may share unobservable patterns in auction design, cost modeling, and project-bundling strategies. To assess how much our predictive performance depends on state-specific patterns, we used a group-aware shuffle split that approximately *balances* the number of observations, rather than the number of states, in each test set. Under this cross-state evaluation, the pooled out-of-fold R^2 is 0.565, a drop

Table 5: Model comparison predicting $\log(1 + \text{funding per location})$. RMSE and MAE are reported in original dollars for tree-ensemble models only; the inverse-log transform amplifies large residuals in linear models, making dollar-space errors unreliable for comparison. Models trained on an 80/20 train–test split.

Family	Model	Test R^2	CV R^2 (mean ± SD)	RMSE ($)	MAE ($)
Linear	OLS	0.5411	0.5219 ± 0.0210	—	—
	Ridge	0.5411	0.5220 ± 0.0210	—	—
	Lasso	0.5408	0.5221 ± 0.0205	—	—
Random Forest	RF Baseline (200 trees)	0.7090	0.6845 ± 0.0144	2,650	**1,537**
	RF Tuned	**0.7187**	**0.6920 ± 0.0145**	2,645	1,546
XGBoost	XGBoost	0.7175	0.6814 ± 0.0160	**2,617**	1,539
GBR	GBR Tuned	0.7037	0.6629 ± 0.0170	2,642	1,580

from $R^2 = 0.712$ under the conventional random split. This reduction suggests that state-specific patterns or feature-outcome relationships do not transfer uniformly across states. As discussed in §4.2, no individual feature used in our model exhibits a high correlation with state (See Appendix D.1), suggesting that any state-related signal may arise from joint combinations of features rather than any single state proxy.

From a policy perspective, however, cross-state extrapolation is not the primary practical use case. Since subgrantee selection is state-specific, practical cost modeling would incorporate existing state-level information rather than predict outcomes for states absent from the training set. Therefore, our original 5-fold model evaluation better represents the model's intended application: given the feature set, predicting generalized APLF for deployment projects.

(2) Within-state modeling. To evaluate how effectively it predicts APLF within each state, we trained and fitted the model on state-specific subsets.[10] Table 7 shows the top-5 performing states in terms of test R^2.[11]

We further explained within-state predictability by comparing the correlation between each within-state R^2 and the feature set. We found that the average per-project satellite and fixed wireless share had the highest correlation, indicating technology composition impacts funding variances, which we explored further in §5.4. The number of projects in each state also correlates with within-state R^2, indicating that larger counts produce stronger R^2 estimates. Thus, within-state R^2 estimates for states with fewer than 50 projects (e.g., NJ, NH) should be interpreted cautiously, as small test-set sizes could produce high evaluation variance.

Table 7: Within-group Random Forest (State)

State	n	CV R^2	test R^2
OR	304	0.9651 ± 0.0164	0.977
NJ	29	0.4127 ± 0.6155	0.930
TN	126	0.8800 ± 0.0473	0.919
WA	374	0.9285 ± 0.0291	0.903
NH	27	0.4854 ± 0.3681	0.893

[10] E.g., OR results are trained and fitted using 313 projects in Oregon.
[11] Full table in Appendix E.1. VT is excluded despite high test $R^2 = 0.954$ due to unstable CV performance.

5.4 Technology-Stratified Modeling

To examine how well the model performs if trained on technology-stratified subsets, we created subsets based on each project's most prevalent technology type.[12] Table 8 summarizes the results. LEO-dominant projects achieved the highest R^2, likely because cost-per-site is more uniform and predictable from fewer variables. HFC-dominant projects are included but not interpreted, since the small sample size is insufficient for a stable RF fit.

Table 8: Within-group Random Forest (Technology)

Group	n	CV R^2	test R^2
61 (LEO)	1882	0.5955 ± 0.0356	0.587
71 (FW)	666	0.3634 ± 0.0371	0.354
50 (fiber)	2548	0.3480 ± 0.0324	0.319
40 (HFC)	67	0.3029 ± 0.2415	0.235

We further examined feature importance in each subset and summarized the results in Appendix E.2. We found that `miles_per_loc` was the most important feature for all technology types except LEO Satellite (61), which is intuitive since fiber miles are uniformly 0 for LEO-dominant projects. For fiber-dominant projects, other top features include rural population density, distance to any nearest served location, distance to the awardee ISP's nearest served location, and percentage of assigned locations with existing presence, suggesting that rurality and existing infrastructure dominate prediction and variability in APLF.

Interestingly, project size (`n_total`) is among the top five features for all except fiber, suggesting that project scale affects APLF. Furthermore, the cost of LEO should be the same throughout the U.S., as satellites are not specific to a region. Thus, any difference in APLF is likely caused by the extent of competition by terrestrial BEAD bidders anticipated by the LEO provider.

6 Discussion

At its core, BEAD's attempt to bring universal internet access illustrates the challenge of minimizing deployment costs while building

[12] E.g, "LEO-dominant projects" are projects with a majority of LEO-assigned locations.

long-term survivability and resilience into broadband connectivity. Modeling BEAD funding decisions surfaces not only cost drivers that policy standards explicitly name—remoteness, population density, topography, and poverty—but also latent features such as the absence of physical infrastructure, market competition, and awardee presence. We contribute to this intersection of infrastructure resilience, telecommunications policy, and network economics in the following ways.

First, we present one of the first **cross-state predictive models of BEAD funding per location**. Tree-based models capture the nonlinear relationship between infrastructure build requirements and funding distribution that linear approaches miss and identify miles per location and technology type as primary predictors, with market competition secondary. This finding echoes Lehr [16] and Jamison [15], which argued that BEAD's original fiber priority inflated cost requirements. The June 2025 RPN removed the fiber-as-priority designation in favor of cost-saving [28]. However, no data exist to test the counterfactual: namely, whether FP drafts before the June 2025 RPN had higher fiber assignments and thus higher APLF. In federal funding programs, technology requirements often stem from performance mandates—such as those outlined in Zhu [34]—and necessarily drive up costs, as Cartesian and ACA Connects [9] have shown, creating a long-lasting tension between performance and cost.

Second, **our tree-based model explains approximately 71.9% of funding variance** using interpretable features: infrastructure deployment metrics, factors from NTIA's cost model, and NBM-derived features. Our model shows that, beyond Broadband, tree-based models may be well suited to policy-driven funding datasets, since program rules often impose thresholds and eligibility cutoffs (e.g., BEAD's high-cost designation and EHCPLT) that create the nonlinear, interaction-heavy structure trees capture natively but linear models cannot. Administrative data aggregated across jurisdictions is also prone to collinearity, as our county-level features illustrate. Tree ensembles' feature subsampling handles this without the coefficient-shrinking compromises regularization requires.

Our finding that a **single engineering metric—fiber miles per location—surpasses multiple named NTIA factors** may generalize beyond broadband: a similar pattern may exist in other infrastructure domains undergoing digital transformation, where policy frameworks similarly rely on composite geographic and demographic factors. If direct engineering metrics dominate in these fields, policy standards could evolve accordingly by anchoring allocation decisions to deployment specifications to improve predictive accuracy and transparency.

7 Limitations & Future Work

7.1 Limitations

Data Limitation. Since not all state broadband offices released their NTIA-approved **final proposal data**, we were only able to run analysis on 49 states and the District of Columbia (See Appendix B.2). U.S. territories were excluded due to their distinct funding structures and fundamentally different deployment characteristics that make them incomparable in a pooled regression. For our dataset, we counted both BSLs and CAIs as locations, since final proposals record them together, and we control for the CAI share of each project with `pct_cai`.

Another data quality limitation comes from FCC's **National Broadband Map (NBM)**, which contains self-reported broadband availability by ISP [10]. Even though the FCC's challenge process acts as a check and balance, any market concentration metrics used in this paper are proxies at best, and we would need tools such as the Broadband Query Tool (BQT) to verify serviceability at each location [22].

7.2 Future Work

Future work could iterate on our data-analytic approach to inform similar questions on policy standards and infrastructure funding.

On granularity, we originally tried training the model on the location-level ($n = \approx 3.70M$), but observed that other than county-level and competitor proxies that differ on the location level—which we later aggregated to project level using weighted means and calculated percentages—other features in our model are largely the same on the project-level. The model also assumes independence across all projects, although similar auction designs may yield some correlations *within* each state—such as project subdivision methodology and cost model application—that our model cannot capture. Our state-grouped cross-validation in §5.3 confirms this: pooled out-of-fold R^2 falls from 0.719 to 0.568. Within-state model performance provides a complementary finding: states with higher average project-level LEO and FW shares are more predictable internally, consistent with the simpler cost structure of non-wired deployments. Future work could utilize a hybrid approach, investigating whether broadband funding patterns generalize across states with similar structural characteristics—such as terrain and technology composition—through cluster- or region-based cross-validation, rather than treating each state as fully independent.

To achieve universal internet access, which is BEAD's goal in the U.S., other crucial questions are open for investigation. On the demand side, even in locations where ISPs won BEAD funding, adoption and affordability may pose hurdles to residents in areas of high rurality and low socioeconomic status [7, 33]. The relationship between fiber internet and artificial intelligence (AI) raises questions about competing labor demands and offers integration opportunities [13]. BEAD's vision of providing internet for all amidst rapid digitization proves to be an important question not only for policy-making and economics, but also for the digital transformation landscape.

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Ethics and Privacy Statement

No human subjects data was collected. All analysis uses publicly available government datasets, with the exception of BSL fabric data, obtained via Columbia University's Tier 4 Research License with CostQuest.

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Appendices are supporting material that has not been peer-reviewed.

A Glossary

Below is a list of definitions that aims to help readers contextualize the BEAD project better.

Broadband Serviceable Locations (BSLs): "a business or residential location in the U.S. at which mass-market internet access service is, or can be, installed"[13]

Eligible Entities: 56 states and territories in the United States that are authorized to receive federal funding.

Subgrantees: entities—including internet service providers (ISPs), local governments, and non-profits—that receive these funds through a competitive process (e.g., auction-style bidding) to deploy broadband[14]

NTIA: The National Telecommunications and Information Administration (NTIA) administers and sets the rules for BEAD across all 56 states and territories. It works with the FCC, using its national broadband map data to determine how much funding each state gets. In order to get the funding, each state must submit its initial proposal, and after it gets approved by the NTIA, states will then submit the final proposal wherein the state shows which exact locations will be funded and which ISP will serve these locations.

Final Proposal (FP): the "'final submission' for the Eligible Entity's BEAD grant funding; [...] should provide an update to the approved Initial Proposal, as modified by the BEAD Restructuring Policy Notice."[15] Includes CSV files:

- **Subgrantees (fp_subgrantees.csv)**: list of subgrantees with allocated funding
- **Deployment Projects (fp_deployment_projects.csv)**: project-level data; each project contains multiple locations; contains BEAD support funding
- **Locations (fp_locations.csv)**: all BEAD-eligible locations that got funding
- **No BEAD Locations (fp_no_BEAD_locations.csv)**: Eligible locations that didn't receive funding, alongside columns such as the reason
- **Community Anchor Institutions (fp_cai.csv)**: "only applicable if the Eligible Entity plans to serve Community Anchor Institution (CAI) locations"

Community Anchor Institution (CAI): a facility that serves as a hub for a community and typically needs reliable broadband to deliver its services. Under BEAD, the statute defines CAIs as the following entities: schools (K-12), libraries, healthcare providers (such as clinics and hospitals), public safety entities (fire, police, EMS), institutions of higher education, public housing organizations, and other community support organizations that facilitate greater use of broadband, especially for vulnerable populations (low-income, unemployed, aged).

[13]https://www.ntia.gov/funding-programs/high-speed-internet-programs/broadband-equity-access-and-deployment-bead-program
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B Data Collection

We collect and store the datasets used for this paper as Comma-Separated Values (CSV) files in our repository or in a BigQuery database on Google Cloud Platform.

B.1 Broadband Data Collection

The National Broadband Map (NBM) is published biannually [10]. Each release reflects ISP filings of service availability as of a reporting date (June 30 or December 31), and it is tied to a specific version of the BSL Fabric, which ISPs use to enumerate the locations they serve. For example, the Version 6 Fabric was used by ISPs to report service as of December 2024 and was released in May 2025. We use the December 2024 as-of data (Fabric v6), the most recent snapshot available when Eligible Entities were drafting their updated final proposals. The FCC provides an API that allows data download for all states. The NBM provides a snapshot as of the release date, with information including location ID, ISPs and their claimed download/upload speed, and census block GEOID.

To compute provider presence at the location level, we performed extensive data cleaning to match the FCC Registration Number (FRN), holding company name, and/or geography served of the subgrantees in each state's final proposal data with the NBM. We created a Provider ID field in the final proposal data for merging.

The BDC was also used to obtain the block GEO ID for each location. Since location IDs get updated with each BDC release, around $17,698$ locations out of $3,386,647$ locations in the final proposal are not found in the December 2024 NBM, which accounts for 0.52% of the data and an even smaller percentage affected, as the dataset was aggregated to project-level.

B.2 Final Proposal Data

Final proposal data is collected manually from each state's broadband office and corroborated with the "Benefit of the Bargain" figures from NTIA's BEAD Public Resources [18]. For example, the reported total funding allocated to each state is matched by summing and rounding the BEAD support column of each state's deployment project CSV. A snippet of the described difference is summarized in Table 9. States with an absolute percentage change < 0.01 in any variable — except for average per-location subgrantee match, since it wasn't considered in this paper—were omitted. As of the paper's submission date, there is no NTIA data available to corroborate the District of Columbia.

States were asked to follow the templates from NTIA [20] when submitting their final proposal for approval. However, we encountered some states with different naming conventions for the columns and had to perform column name normalization, data type matching, and dataset encoding normalization before uploading all datasets to BigQuery.

Table 9: Difference between NTIA reported figures and final proposal data available from State Broadband Offices, sorted by percent change in total BEAD support

state	Δbs (%)*	Δn_fttp (%)	Δn_hfc (%)	Δn_lfw (%)†	Δn_fw (%)	Δn_leo (%)	Δn_total (%)	ΔAPLF (%)	ΔAPLSM (%)
RI	-34.53	0.00	N/A	0.00	N/A	0.00	0.00	-34.52	-18.92
PA	-10.05	-10.55	40.18	-0.91	N/A	29.60	-0.15	-9.90	-5.82
GA	-1.64	0.03	0.05	N/A	N/A	0.55	0.09	-1.74	0.37
AL	1.12	0.95	0.00	N/A	N/A	5.48	2.00	-0.87	1.78
IN	-0.63	-6.91	-3.02	-7.99	N/A	-6.68	-6.93	6.79	5.67
MO	-0.23	-4.16	-6.82	N/A	35.35	-19.65	-4.73	4.71	5.94
DE	-0.13	-6.84	49.83	N/A	N/A	N/A	0.23	-0.38	0.01
SD	-0.02	-0.17	N/A	0.00	N/A	2.85	0.97	-0.98	-0.94
IL	0.00	0.00	-40.12	N/A	0.00	0.00	-0.09	0.09	0.14
AR	0.00	0.14	0.00	0.00	N/A	-0.65	0.00	0.00	0.00
DC	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

* Δ represents the percentage change from the state-published FP to the NTIA-published figures. For example, a negative value means the NTIA-approved figure is lower than the one published on the state's website as of September 18th, 2026. The full forms for unseen acronyms are as follows:
  - **bs**: total bead support
  - **APLSM**: average per-location subgrantee match

† Throughout the paper, all fixed wireless technology (70, 71, 72) are grouped as one, but since NTIA's summary statistics differentiates between licensed FW (LFW) and FW, we follow its convention.

C Additional Figures

The full bar charts of all available states for Figures 4 and 5 are shown here.

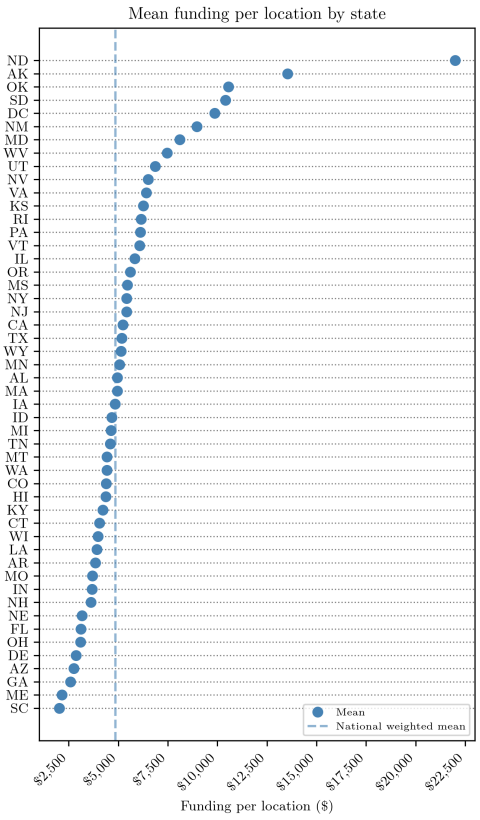


Figure 8: Mean BEAD funding per location by state.

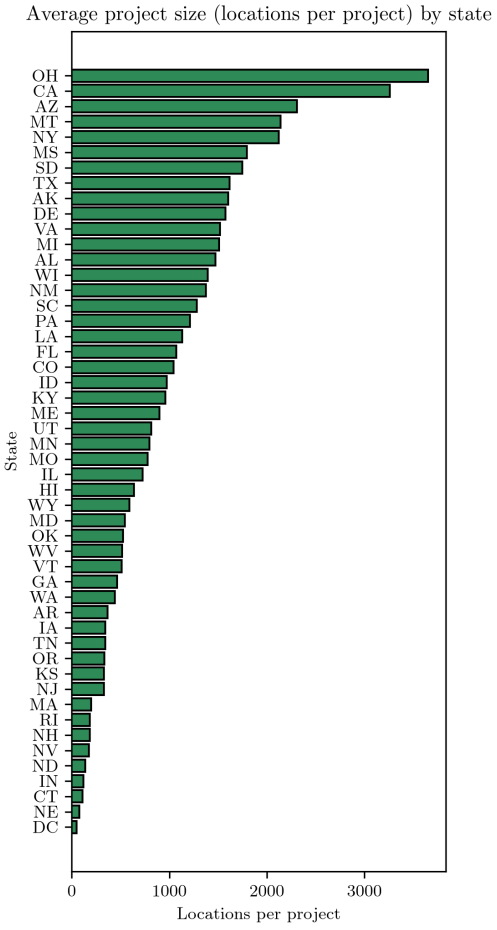


Figure 9: Mean project size by state.

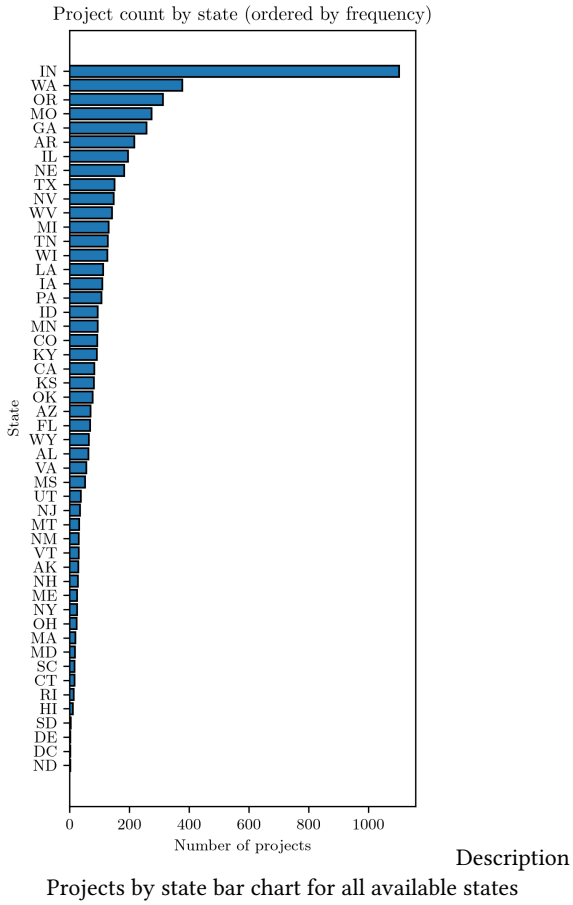


Figure 10: Count of the number of projects per state.

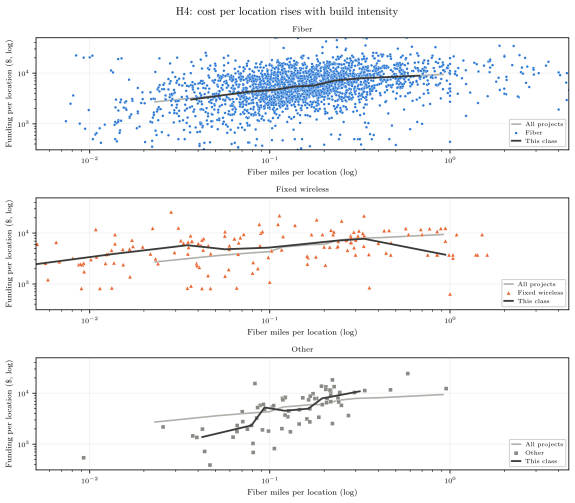


Figure 11: Funding per location versus fiber miles per location across technology classes.

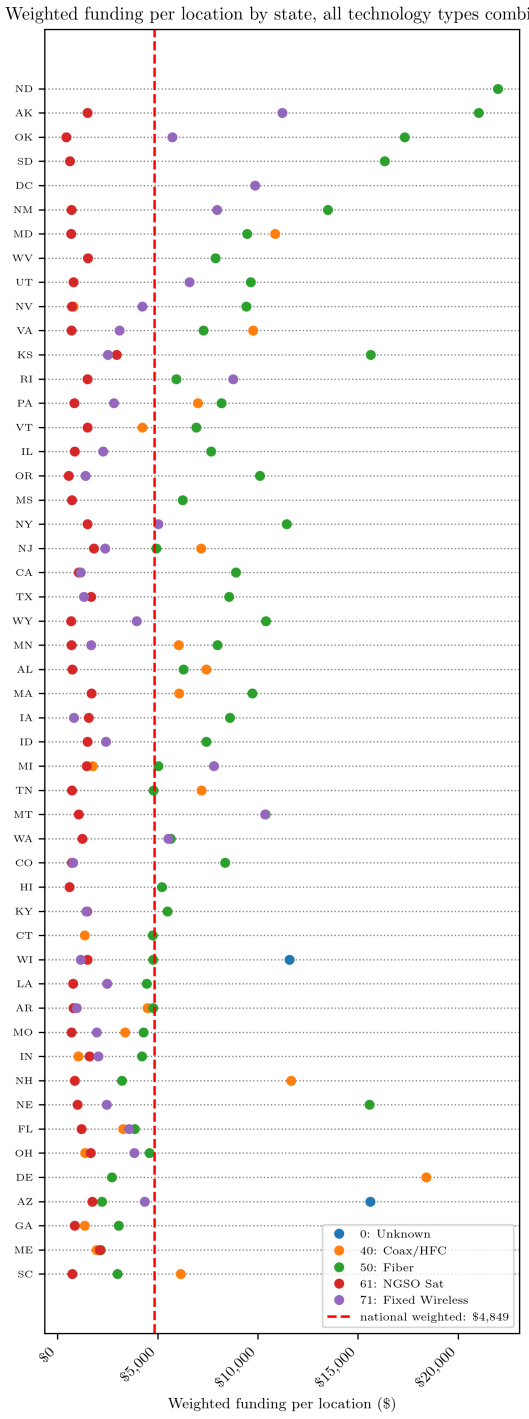


Figure 12: Mean BEAD funding per location by state stratified by technology.

D Additional Information - Input Features

D.1 Feature ANOVA

Table 10 shows how much each feature is related to the state variable.

Table 10: % variance in each feature explained by state

feature	η^2	% variance by state
rur_popden	0.356931	35.7
poverty_rate_wmean	0.318397	31.8
no_prior_presence_frac	0.303752	30.4
frac_tech_61	0.297510	29.8
pct_aerial	0.296819	29.7
pct_cai	0.290484	29.0
frac_tech_50	0.256701	25.7
frac_q_comp	0.235684	23.6
cou_popden	0.227835	22.8
prov_coverage	0.225523	22.6
frac_frac_presence	0.214730	21.5
frac_tech_71	0.174500	17.4
rucc_code_wmean	0.145350	14.5
nearest_own_mi	0.120164	12.0
frac_q_wired	0.117011	11.7
frac_tech_40	0.107258	10.7
nearest_any_mi	0.083671	8.4
n_total	0.058095	5.8
miles_per_loc	0.003490	0.3

D.2 Feature Importance

Table 11 shows the full table of all the features' relative importance, ranked in descending order based on their SHAP values.

Table 11: Normalized feature importance from RF Gini and SHAP

Feature	RF Gini	SHAP
miles_per_loc	0.241	0.275
frac_tech_61 (LEO)	0.099	0.174
frac_tech_50 (Fiber)	0.126	0.142
nearest_own_mi	0.063	0.061
rur_popden	0.049	0.045
frac_presence	0.039	0.044
nearest_any_mi	0.045	0.043
isp_coverage	0.053	0.034
poverty_rate	0.037	0.028
n_total	0.048	0.023
frac_q_wired	0.025	0.023
frac_tech_71 (FW)	0.024	0.020
pct_aerial	0.026	0.018
cou_popden	0.039	0.017
frac_q_comp	0.030	0.016
rucc_code	0.025	0.014
no_prior_presence	0.012	0.013
pct_cai	0.014	0.009
frac_tech_40 (HFC)	0.002	0.001

D.3 Feature Ablation

Feature ablation showing how removing certain features affects test R^2. We document the change in R^2 when we remove an individual feature. A positive change is noted under the double lines; however, we did not remove the features, as they improve overall model performance in conjunction with other features.

Table 12: Feature ablation results. Baseline Test R^2 = 0.7187.

Feature	Test R^2	ΔR^2	RMSE ($)	MAE ($)
miles_per_loc	0.6771	−0.0416	2,858	1,693
rur_popden	0.7000	−0.0187	2,702	1,588
pct_cai	0.7089	−0.0098	2,665	1,564
pct_aerial	0.7103	−0.0084	2,686	1,576
n_total	0.7111	−0.0076	2,653	1,557
cou_popden	0.7129	−0.0057	2,658	1,559
poverty_rate	0.7138	−0.0049	2,646	1,552
rucc_code	0.7139	−0.0048	2,654	1,553
no_prior_presence	0.7145	−0.0042	2,661	1,555
nearest_any_mi	0.7163	−0.0024	2,649	1,550
frac_presence	0.7164	−0.0023	2,648	1,549
nearest_own_mi	0.7173	−0.0014	2,653	1,557
frac_tech_40	0.7176	−0.0011	2,646	1,551
frac_tech_71	0.7179	−0.0008	2,645	1,544
frac_tech_61	0.7182	−0.0005	2,647	1,550
isp_coverage	0.7184	−0.0003	2,646	1,548
frac_q_wired	0.7211	+0.0024	2,629	1,540
frac_tech_50	0.7200	+0.0014	2,638	1,543
frac_q_comp	0.7196	+0.0009	2,647	1,543

D.4 Feature Motivation

Table 13 documents the motivation and supporting literature for each of the 19 input features used in the final Random Forest model. Features are grouped by category matching Table 3.

Table 13: Motivation and literature for each input feature in the final model.

Feature	Motivation	Supporting Literature / Rationale
Project-level		
n_total	Project size (location count)	Larger projects spread fixed mobilization and engineering overhead across more locations, reducing marginal cost per location. Included to capture economies of scale within a project.
frac_tech_40	Cable share within project	Cable (HFC) projects receive just as much median per-location funding ($5,531) as fiber ($5,865), though they represent around 1% of projects (H5, § 3.3). Including the cable share captures this distinct cost tier. One-hot encoding avoids imposing an ordinal cost ordering across the four technology categories.
frac_tech_50	Fiber (FTTP) share within project	Fiber projects receive significantly higher per-location funding than fixed wireless (median $6,804 vs. $3,518, H2 in § 3.3). Jamison [15] critiques NTIA's fiber prioritization as a policy-driven cost driver; Lehr [16] argues fiber mandates raise deployment costs relative to fixed wireless alternatives. Cartesian and ACA Connects [9] documents that per-location fiber costs rise sharply in the highest-cost deciles, from $2,700 to over $95,000.
frac_tech_61	LEO satellite share within project	LEO satellite projects fall in the lowest-cost tier ($1,000–$2,700 per location, H1 in § 3.3). The SHAP beeswarm (Figure 7) shows that a higher LEO share within a project consistently drives predicted APLF downward, reflecting the lower infrastructure requirements of non-terrestrial deployment.
frac_tech_71	Fixed wireless share within project	Fixed wireless median funding ($3,518) is roughly half that of fiber (H2, § 3.3), reflecting lower infrastructure requirements. Including the fixed wireless share alongside the other three technology fractions completes the one-hot decomposition.
miles_per_loc	Build intensity: total fiber miles per funded location	The strongest predictor in both Gini importance (0.201) and SHAP (0.250). Dispersed rural locations require more fiber per location than dense clusters, directly reflecting deployment capital requirements [9]. The SHAP beeswarm (Figure 7) shows a consistent positive monotonic effect: longer miles per location substantially increases predicted APLF across all projects.
pct_aerial	Fraction of fiber miles deployed aerially (vs. buried)	Median per-foot deployment cost is roughly twice as high underground as aerial ($18.00/ft vs.$8.00/ft in 2025), and underground costs are far more sensitive to terrain and region [14]. This is consistent with H3 (§ 3.3), which found no per-location difference between burial methods in isolation: per-foot and per-location costs diverge with density, so the aerial share carries signal only in conjunction with miles per location, holding technology and terrain constant.
pct_cai	Fraction of funded locations that are Community Anchor Institutions (CAIs)	BEAD defines eligible CAIs (schools, libraries, health facilities) as a distinct location class subject to a 1 Gbps symmetric service standard, versus 100/20 Mbps for other broadband-serviceable locations, and permits entities to fund them only after ensuring coverage of all unserved and underserved locations [26]. Including the CAI share tests whether a higher concentration of institutional locations shifts per-location funding.
Competition		
frac_q_comp	Fraction of locations with any qualifying competitor: an ISP other than the awardee that reports terrestrial (non-satellite) service at 100/20 Mbps or better in the NBM	Franchise bidding disciplines subsidy levels only when the field of bidders is thick and credible [32], and in BEAD a larger set of existing providers both raises bidding pressure and narrows the scope for post-award subsidy-seeking [23].
frac_q_wired	Fraction of locations with a qualifying *wired* competitor (cable or fiber): a qualifying competitor (defined above) whose technology is cable (40) or fiber (50)	Separates satellite-driven signal from genuinely competitive terrestrial markets, since a qualifying wired competitor implies existing physical plant that lowers expected build cost (reused conduit runs roughly 20% below new-conduit installation [14]) whereas satellite presence carries no comparable cost or competitive implication [17].

Continued on next page

Table 13 – continued from previous page

Feature	Motivation	Supporting Literature / Rationale
Awardee ISP		
isp_coverage	Awardee's mean census-block coverage (%) across funded locations	Blocks where the awardee already reports broad coverage are blocks where its existing network runs closer to the funded locations, which cost models treat as a key driver of build cost [12]; the expected effect is still ambiguous, since the locations left needing funding in a well-covered block are the ones the provider chose not to build to before, and may be the hardest to reach.
frac_presence	Fraction of funded locations where awardee ISP is already present in the NBM	Location-level incumbency: because BEAD funds only unserved and underserved locations, awardee presence in the NBM [10] implies the award upgrades the provider's own existing service rather than building to a location with no plant, a materially cheaper proposition. Rosston and Wallsten [23] note that bidders may nonetheless bid below sustainable cost in anticipation of later subsidy support.
nearest_own_mi	Mean distance (miles) to the nearest location the *same* ISP already serves	Operationalizes extension vs. new-build cost: a provider expanding from locations it already serves faces lower marginal construction cost than one deploying where it has no plant, with distance proxying the length of distribution and last-mile construction required.
nearest_any_mi	Mean distance (miles) to the nearest location served by *any* ISP	Captures spatial remoteness from existing broadband infrastructure of any kind. Remote location is one of the statutory factors defining high-cost areas under BEAD [1, 27]. A longer distance to the nearest served location of any provider suggests that the location is in a genuinely underserved corridor where infrastructure of any kind is sparse.
no_prior_presence	Fraction of funded locations with no existing same-ISP service anywhere nearby	Identifies fully greenfield projects where the awardee has no footprint to build from. Pure greenfield deployments carry higher fixed costs (permitting, right-of-way, backhaul) than incremental extensions, motivating higher per-location funding.
County-level (location-weighted means)		
rur_popden	Rural population density (persons/mi^2 of rural land area), county weighted mean	Lack of population density is one of the statutory factors defining high-cost areas under BEAD [1, 27], with rural land area delineated per U.S. Census Bureau [29]. FCC estimates reported in Lehr [16] document that FTTP costs rise steeply as density falls, reporting roughly $1,822 per location for the first 98% of unserved households against $89,286 for the final 2%.
cou_popden	County-wide population density (persons/mi^2), county weighted mean	Two density measures are included because rural land density and overall county density capture distinct dimensions of deployment economics.
rucc_code	Rural-Urban Continuum Code (RUCC), county weighted mean	The RUCC (1 = metro counties of 1M+; 9 = completely rural, nonadjacent) is USDA's standard rurality classification [31], and remote location is one of the statutory factors defining high-cost areas under BEAD [1, 27]. We treat the ordinal scale as cardinal when averaging across counties spanned by a project.
poverty_rate	County poverty rate (ACS 5-year estimate), location-weighted mean	High poverty rate is one of the statutory factors defining high-cost areas [1, 27], and the NOFO further requires a strong preference for high-poverty and persistent-poverty areas when funds are insufficient, and restricts the early 20% funding release to high-poverty projects [27]. Measured from ACS 5-year estimates [30].

E Other ablation details

E.1 State Ablation

The full table for state-stratified modeling is presented in Table 14. States with negative mean CV R^2 or test R^2 are excluded due to unstable fit.

Table 14: Within-group Random Forest (State)

Group	n	CV R^2	test R^2
OR	304	0.9651 ± 0.0164	0.977
NJ	29	0.4127 ± 0.6155	0.930
TN	126	0.8800 ± 0.0473	0.919
WA	374	0.9285 ± 0.0291	0.903
NH	27	0.4854 ± 0.3681	0.893
NV	106	0.7348 ± 0.1563	0.864
MA	19	0.3287 ± 0.6638	0.863
IA	106	0.6591 ± 0.0757	0.799
KS	71	0.5725 ± 0.1998	0.776
WI	123	0.4889 ± 0.1211	0.741
AL	63	0.3732 ± 0.4839	0.736
GA	258	0.6952 ± 0.1114	0.732
VA	55	0.4080 ± 0.2819	0.720
OK	60	0.2797 ± 0.3357	0.706
LA	108	0.2267 ± 0.4842	0.696
IN	1030	0.5324 ± 0.0545	0.663
WV	140	0.8155 ± 0.0699	0.655
TX	150	0.6524 ± 0.1576	0.655
MO	265	0.4572 ± 0.1062	0.632
CO	92	0.6232 ± 0.0920	0.624
MS	51	0.1013 ± 0.6775	0.590
IL	194	0.4697 ± 0.0536	0.587
NE	175	0.4262 ± 0.2582	0.555
KY	91	0.0341 ± 0.2187	0.541
WY	56	0.6462 ± 0.0791	0.523
AR	211	0.7838 ± 0.1169	0.521
MN	90	0.2425 ± 0.1864	0.443
FL	69	0.0464 ± 0.3569	0.394
MI	130	0.4198 ± 0.3696	0.369
CA	76	0.6192 ± 0.1417	0.296
ID	86	0.2367 ± 0.1441	0.268
AZ	70	0.0176 ± 0.2780	0.225
PA	95	0.1770 ± 0.2280	0.047

Table 15 shows the pairwise correlation between the given feature and the within-group test R^2. It only includes features with a positive correlation.

Table 15: Correlation between within-group test R^2 and feature

feature	r^2
cable_share	0.002779
mean_popden	0.052912
funding_std	0.094840
n_projects	0.105990
fw_share	0.118194
sat_share	0.182312

E.2 Technology Ablation Feature Importance

Table 16: Top 5 normalized feature importance from RF Gini and SHAP for HFC-dominant projects

Feature	RF Gini	SHAP
miles_per_loc	0.151	0.215
cou_popden	0.076	0.108
frac_q_comp	0.094	0.094
n_total	0.097	0.088
pct_cai	0.060	0.082

Table 17: Top 5 normalized feature importance from RF Gini and SHAP for fiber-dominant projects

Feature	RF Gini	SHAP
miles_per_loc	0.210	0.328
rur_popden	0.092	0.128
nearest_own_mi	0.069	0.075
nearest_any_mi	0.063	0.069
frac_presence	0.050	0.059

Table 18: Top 5 normalized feature importance from RF Gini and SHAP for LEO-dominant projects

Feature	RF Gini	SHAP
rur_popden	0.198	0.190
frac_presence	0.096	0.138
n_total	0.122	0.136
isp_coverage	0.072	0.109
cou_popden	0.138	0.081

Table 19: Top 5 normalized feature importance from RF Gini and SHAP for FW-dominant projects

Feature	RF Gini	SHAP
miles_per_loc	0.090	0.160
n_total	0.110	0.095
frac_q_comp	0.078	0.092
frac_tech_50 (Fiber)	0.050	0.084
frac_tech_71 (FW)	0.050	0.082

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